

ABG Interpretation Workshop

Participant Discussion and Answer Booklet - 13 Cases

The Unified 10-Step Approach

Step	Tool / key question
1. Clinical history	What does the story alone suggest?
2. Internal validity	$[H^+] = 24 \times PaCO_2 / HCO_3^-$ - does it match pH? Arterial?
3. pH	Acidaemia, alkalaemia, or normal? A normal pH does not mean a normal gas.
4. Primary disorder	Does $PaCO_2$ or HCO_3^- explain the pH direction?
5. Expected compensation	Metabolic acidosis: $PaCO_2 = 1.5 \times HCO_3^- + 8 \pm 2$. Metabolic alkalosis: $PaCO_2 = 0.7 \times HCO_3^- + 21 \pm 2$.
6. Anion gap	$Na^+ - (Cl^- + HCO_3^-)$; add 2.5 per 1 g/dL albumin below 4.0.
7. Delta ratio	$(AG - 12)/(24 - HCO_3^-)$ - only if AG raised.
8. Oxygenation & integration	P/F, OI, CaO_2 , $ScvO_2$, MetHb. Does the story fit the child?
9. SID / Stewart analysis	$SID = Na^+ - Cl^-$ (normal ~36). Low = chloride-driven acidosis. High = chloride-loss alkalosis. $Atot = \text{albumin} + \text{phosphate}$. Unmeasured anions = $AG - (\text{lactate} + \text{ketones})$.
10. Post-intervention re-evaluation	Has the mechanism changed? Recalculate AG, SID, $Atot$.

Case 1 - DKA

Vignette. 10-year-old girl, 2 weeks of polyuria and 3 days of vomiting and abdominal pain. Acidotic breathing present, haemodynamically stable, GCS 11.

Initial gas

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.08	14	4	36	132
K	Cl	Albumin	Glucose	b-OHB
4.5	98	4.0	480	5.8

Systematic interpretation - answers

Step	Answer
1. Clinical history	2 weeks of polyuria (osmotic diuresis) with vomiting and abdominal pain, Kussmaul breathing, GCS 11 - classical new-onset DKA with early obtundation.
2. Internal validity	$[H^+] = 24 \times 14 / 4 = 84 \text{ nmol/L} \rightarrow \text{pH } 7.08$ - matches. PaO ₂ 36 suggests a venous sample.
3. pH	7.08 - severe acidaemia.
4. Primary disorder	HCO ₃ 4 (low) with acidaemia $\rightarrow$ metabolic acidosis.
5. Expected compensation	Winter's: $\text{PaCO}_2 = 1.5 \times 4 + 8 = 14 \pm 2$. Actual 14 - appropriate.
6. Anion gap	$\text{AG} = 132 - (98 + 4) = 30$ (markedly raised). Albumin normal - no correction needed.
7. Delta ratio	$(30 - 12) / (24 - 4) = 0.9 \rightarrow$ pure HAGMA.
8. Oxygenation & integration	Severe HAGMA with maximal appropriate respiratory compensation; AG 30 fully explained by ketoacids (b-OHB 5.8). Severe DKA.

SID / Stewart - answers

Tool	Value	Interpretation
SID	$132 - 98 = 34$	Low-normal - mild chloride-driven component already present.
Atot	Albumin 4.0	Normal.
Unmeasured anions	$30 - 5.8 = 24.2$	Markedly raised - ketoacids dominate.
Mechanism	-	Unmeasured anions, with mild chloride-driven component.

Case-specific questions

1. Calculate the corrected serum sodium.
2. What do the glucose and b-OHB values suggest?

Answers

1. $\text{Na}^+ + 1.6 \times [(480 - 100)/100] \sim 138$. True hyponatraemia is dilutional from hyperglycaemia.
2. Hyperglycaemic ketosis driving the gap - severe DKA.

Post-intervention gas

Intervention. 10 mL/kg NS bolus, insulin infusion, NS + KCl.

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.25	26	11	34	142
K	Cl	Albumin	Glucose	b-OHB
3.4	119	4.0	230	0.6

Post-intervention interpretation

Step	Answer
2. Internal validity	[H ⁺] ~56 nmol/L - internally valid.
3. pH	7.25 - improving.
4. Primary disorder	HCO ₃ 11 -> metabolic acidosis.
5. Expected compensation	Winter's: PaCO ₂ = 1.5 x 11 + 8 = 24.5 +/-2. Actual 26 - appropriate.
6. Anion gap	AG = 142 - (119 + 11) = 12 (normal - HAGMA resolved).
7. Delta ratio	(12 - 12)/(24 - 11) = 0 -> pure NAGMA.
9. SID / Stewart	SID = 142 - 119 = 23 (markedly low - chloride-driven). Unmeasured anions = 12 - 0.6 = 11.4 (normal). Mechanism changed: ketoacids cleared, chloride load now driving.

Case-specific questions

1. Compare this ABG with the initial ABG. What has changed?
2. Why is acidosis persisting despite improvement in glucose and b-OHB?
3. Calculate the corrected serum sodium.

Answers

1. pH improved (7.08->7.25); PaCO₂ rose (14->26); HCO₃ rose modestly (4->11); glucose fell (480->230); b-OHB fell (5.8->0.6); Na rose (132->142); K fell (4.5->3.4); Cl rose sharply (98->119); AG normalised (30->12). The acid has changed character.
2. Ketoacidosis resolved. Persisting low HCO₃ is now hyperchloraemic NAGMA from saline chloride load - expected, not treatment failure.
3. $142 + 1.6 \times [(230 - 100)/100] \sim 144$.

Take-home

Recalculate the anion gap on every repeat gas - a persistently low HCO₃ after treatment can mean the acid has changed character.

Case 2 - Pyloric stenosis

Vignette. 6-week-old first-born boy with 10 days of progressively forceful, non-bilious vomiting after every feed. Sunken eyes, lethargic but arousable, still hungry after vomiting.

Initial gas

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.52	48	38	34	132
K	Cl	Albumin	Glucose	Urine Cl
3.0	82	4.0	90	<15

Systematic interpretation - answers

Step	Answer
1. Clinical history	Classic for hypertrophic pyloric stenosis: forceful non-bilious vomiting after every feed, still hungry, with dehydration.
2. Internal validity	$[H^+] = 24 \times 48 / 38 = 30 \rightarrow$ pH 7.52 - matches. PaO ₂ 34 suggests venous.
3. pH	7.52 - alkalaemia.
4. Primary disorder	HCO ₃ 38 (raised) with alkalaemia $\rightarrow$ metabolic alkalosis.
5. Expected compensation	$PaCO_2 = 0.7 \times 38 + 21 = 47.6 \pm 2$ (45.6-49.6). Actual 48 - appropriate.
6. Anion gap	$AG = 132 - (82 + 38) = 12$ (normal).
7. Delta ratio	Not applicable - no HAGMA.
8. Oxygenation & integration	Hypochloraemic, hypokalaemic metabolic alkalosis with appropriate respiratory compensation; low urine Cl confirms chloride-responsive process.

SID / Stewart - answers

Tool	Value	Interpretation
SID	$132 - 82 = 50$	Markedly high - chloride-loss alkalosis.
Atot	Albumin 4.0	Normal.
Unmeasured anions	$AG\ 12 - 0 = 12$	Normal.
Mechanism	-	Loss of gastric HCl has removed chloride, raising SID.

Case-specific questions

1. What is the significance of the low urine chloride?
2. What is the likely clinical diagnosis?

Answers

1. Urine Cl < 15-20 mmol/L confirms chloride-responsive metabolic alkalosis: kidney appropriately conserving chloride. Alkalosis will correct with chloride-rich fluid. Contrast with chloride-resistant causes (Bartter/Gitelman) where urine Cl is inappropriately high.
2. Hypertrophic pyloric stenosis - persistent vomiting of gastric HCl and volume depletion produce hypochloraemic, hypokalaemic metabolic alkalosis; secondary hyperaldosteronism drives further renal H⁺ and K⁺ loss.

Post-intervention gas

Intervention. IV 0.9% saline + KCl, correct dehydration and hypokalaemia, then pyloromyotomy.

pH	PaCO ₂	HCO ₃	Na	K
7.45	40	27	138	4.0

Cl	Albumin	Urine Cl
100	4.0	>20

Post-intervention interpretation

Step	Answer
9. SID / Stewart	SID = $138 - 100 = 38$ (normalising - chloride repletion working). Urine Cl now >20 confirms kidney no longer conserving chloride.

Take-home

A urine chloride confirms whether a metabolic alkalosis will respond to chloride repletion. SID tracks the response.

Case 3 - Gastroenteritis

Vignette. 8-month-old girl with 2 days of fever and large-volume loose stools. Sunken eyes, delayed skin turgor, prolonged capillary refill time.

Initial gas

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.22	27	11	34	129
K	Cl	Albumin	Glucose	Lactate
3.2	112	3.5	85	1

Systematic interpretation - answers

Step	Answer
1. Clinical history	Acute gastroenteritis with large-volume diarrhoea and significant dehydration.
2. Internal validity	$[H^+] = 24 \times 27 / 11 = 59 \rightarrow$ pH 7.23 - matches. PaO ₂ 34 suggests venous.
3. pH	7.22 - acidaemia.
4. Primary disorder	HCO ₃ 11 (low) $\rightarrow$ metabolic acidosis.
5. Expected compensation	Winter's: $PaCO_2 = 1.5 \times 11 + 8 = 24.5 \pm 2$ (22.5-26.5). Actual 27 is at upper edge - borderline compensation.
6. Anion gap	$AG = 129 - (112 + 11) = 6$ (low/normal). Corrected $AG = 6 + 2.5 \times (4 - 3.5) = 7.25$ - still normal.
7. Delta ratio	Not applicable - no raised AG.
8. Oxygenation & integration	NAGMA (hyperchloraemic) from stool bicarbonate loss; normal lactate argues against hypoperfusion.

SID / Stewart - answers

Tool	Value	Interpretation
SID	$129 - 112 = 17$	Markedly low - severe chloride-driven acidosis.
Atot	Albumin 3.5	Mildly low - slightly alkalinising.
Unmeasured anions	$AG\ 6 - 1 = 5$	Normal.
Mechanism	-	Chloride-driven NAGMA.

Case-specific questions

1. What is the likely cause of the metabolic acidosis in this child?
2. How do the chloride and lactate values help you identify the mechanism?

Answers

1. Loss of bicarbonate-rich fluid in large-volume diarrhoea $\rightarrow$ hyperchloraemic NAGMA.
2. Chloride elevated (112) as it replaces lost bicarbonate; normal lactate (1) argues against significant hypoperfusion/shock.

Post-intervention gas

Intervention. IV isotonic fluids, correct dehydration.

pH	PaCO ₂	HCO ₃	Na	K
7.35	36	20	135	4.0
Cl	Albumin	Lactate		
105	3.5	1		

Post-intervention interpretation

Step	Answer
6. Anion gap	$AG = 135 - (105 + 20) = 10$ - normal.
9. SID / Stewart	$SID = 135 - 105 = 30$ (improving).

Take-home

A normal-anion-gap acidosis with a normal lactate points to GI bicarbonate loss, not hypoperfusion. SID confirms the chloride-driven mechanism.

Case 4 - Suspected IEM

Vignette. 8-month-old girl with 1 day of fever and vomiting. Effortless tachypnoea, haemodynamically stable, lethargic. History of similar episodes in the past.

Initial gas

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.24	38	16	38	135
K	Cl	Albumin	Glucose	Lactate
4	107	2	85	3

Systematic interpretation - answers

Step	Answer
1. Clinical history	Short acute illness in an infant with history of similar prior episodes - recurrence is the key word.
2. Internal validity	$[H^+] = 24 \times 38 / 16 = 57 \rightarrow$ pH 7.24 - matches.
3. pH	7.24 - acidaemia.
4. Primary disorder	HCO ₃ 16 (low) $\rightarrow$ metabolic acidosis.
5. Expected compensation	Winter's: $PaCO_2 = 1.5 \times 16 + 8 = 32 \pm 2$ (30-34). Actual 38 - higher than expected $\rightarrow$ additional respiratory acidosis.
6. Anion gap	Raw AG = $135 - (107 + 16) = 12$ (apparently normal). Albumin 2.0: corrected AG = $12 + 2.5 \times (4 - 2) = 17$ (clearly raised).
7. Delta ratio	Using corrected AG: $(17 - 12) / (24 - 16) = 5/8 = 0.625 \rightarrow$ mixed HAGMA + NAGMA.
8. Oxygenation & integration	Concealed HAGMA unmasked only after albumin correction, with inadequate respiratory compensation, in child with recurrent episodes $\rightarrow$ suspect IEM.

SID / Stewart - answers

Tool	Value	Interpretation
SID	$135 - 107 = 28$	Low - chloride-driven component.
Atot	Albumin 2.0	Markedly low - alkalinising, masking HAGMA.
Unmeasured anions	$17 - 3 = 14$	Raised.
Mechanism	-	Three forces: low SID, low Atot (masking), unmeasured anions.

Case-specific questions

1. Calculate the albumin-corrected anion gap.
2. Is there an additional respiratory disorder?
3. What is the significance of recurrent similar episodes in this clinical setting?

Answers

1. $12 + 2.5 \times (4.0 - 2.0) = 17$ mEq/L. Raw gap of 12 looked normal; low albumin was masking a real HAGMA.
2. Yes - Winter's predicts PaCO₂ 30-34; actual 38 is higher $\rightarrow$ inadequate compensation (superimposed respiratory acidosis).
3. Recurrent metabolic decompensation triggered by intercurrent illness raises strong suspicion for underlying IEM (organic acidaemia or fatty acid oxidation defect). Prompt metabolic work-up: ammonia, lactate, urine organic acids, plasma amino acids, acylcarnitine - the critical sample.

Post-intervention gas

Intervention. Critical sample, dextrose, stop protein.

pH	PaCO ₂	HCO ₃	Na	K
7.35	35	20	138	4.2
Cl	Albumin	Glucose	Lactate	
105	2.5	100	2	

Post-intervention interpretation

Step	Answer
9. SID / Stewart	SID = 138 - 105 = 33 (normalising). Atot still low (albumin 2.5) - still masking. Unmeasured anions = corrected AG ($20 + 2.5 \times 1.5 = 23.75$) - 2 = 21.75 (still raised).

Take-home

Correct the anion gap for albumin as a reflex, every time - it is most consequential exactly when the raw number looks reassuring. Also check SID.

Case 5 - Salicylate poisoning

Vignette. 15-year-old boy with tinnitus, fast breathing, vomiting, and confusion after intentional ingestion.

Initial gas

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.42	18	11	108	140
K	Cl	Albumin	Glucose	Lactate
4.2	102	4.1	90	3.2

Systematic interpretation - answers

Step	Answer
1. Clinical history	Tinnitus, hyperventilation, vomiting, confusion after intentional ingestion in a teenager - tinnitus is the most specific clue.
2. Internal validity	$[H^+] = 24 \times 18 / 11 = 39 \rightarrow$ pH 7.41 - matches 7.42. Plausible arterial sample.
3. pH	7.42 - essentially normal (the trap).
4. Primary disorder	Two markedly abnormal primary values (PaCO ₂ 18, HCO ₃ 11) with near-normal pH - mixed disorder.
5. Expected compensation	If HCO ₃ 11 were the only problem, Winter's predicts PaCO ₂ 24.5 +/-2. Actual PaCO ₂ 18 is far below that - separate primary respiratory alkalosis.
6. Anion gap	$AG = 140 - (102 + 11) = 27$ (markedly raised). Lactate only partly explains.
7. Delta ratio	$(27 - 12)/(24 - 11) = 1.15 \rightarrow$ HAGMA.
8. Oxygenation & integration	Primary respiratory alkalosis + primary HAGMA, opposing pH effects cancelling to near-normal pH - classic salicylate poisoning.

SID / Stewart - answers

Tool	Value	Interpretation
SID	$140 - 102 = 38$	High-normal - no chloride-driven component.
Atot	Albumin 4.1	Normal.
Unmeasured anions	$27 - 3.2 = 23.8$	Markedly raised - salicylate metabolites + ketones.
Mechanism	-	Unmeasured anions + primary respiratory alkalosis.

Case-specific questions

1. A near-normal pH can conceal a mixed disorder. What disorders are present here?
2. What poisoning is suggested by the clinical history and ABG pattern?

Answers

1. Primary respiratory alkalosis (PaCO₂ far lower than metabolic compensation alone would predict - direct salicylate stimulation of central respiratory centre) + primary HAGMA (AG 27, from lactate, ketosis, salicylate organic acids).
2. Salicylate (aspirin) poisoning - classic triad of tinnitus, hyperventilation, and mixed respiratory-alkalosis/HAGMA pattern after intentional ingestion in an adolescent.

Post-intervention gas

Intervention. Activated charcoal, IV fluids, sodium bicarbonate, alkalinise urine, consider dialysis.

pH	PaCO ₂	HCO ₃	Na	K
7.48	30	22	142	3.8

Cl	Albumin	Lactate
105	4.1	1.5

Post-intervention interpretation

Step	Answer
9. SID / Stewart	SID = $142 - 105 = 37$ (normal). Unmeasured anions = AG ($142 - 105 - 22 = 15$) - $1.5 = 13.5$ (improving).

Take-home

A near-normal pH never means a normal gas - check PaCO₂ and HCO₃ against each other before you trust it. SID helps distinguish chloride-driven from unmeasured-anion-driven acidosis.

Case 6 - Post-arrest hanging

Vignette. 5-year-old boy, 20 kg, accidental hanging. Post-arrest, ROSC after 15 min CPR, ventilated. PEEP 6, PIP 16, RR 20, I:E 1:2, FiO2 100%, TV 80 mL, MAP 17.

Initial gas

pH	PaCO ₂	HCO ₃	PaO ₂	Na
6.88	65	12	70	138
K	Cl	Albumin	Glucose	Lactate
6.2	98	4.1	210	3.2

Systematic interpretation - answers

Step	Answer
1. Clinical history	Post-cardiac-arrest after prolonged CPR for hanging - expect hypoxic-ischaemic metabolic acidosis and possible inadequate ventilation.
2. Internal validity	$[H^+] = 24 \times 65 / 12 = 130 \rightarrow \text{pH } 6.89$ - matches 6.88.
3. pH	6.88 - profound, life-threatening acidaemia.
4. Primary disorder	Both HCO ₃ 12 and PaCO ₂ 65 abnormal in acidifying direction - combined metabolic + respiratory acidosis.
5. Expected compensation	If pure metabolic, Winter's predicts PaCO ₂ 26 $\pm$ 2. Actual 65 far above - large separate primary respiratory acidosis.
6. Anion gap	$\text{AG} = 138 - (98 + 12) = 28$ (markedly raised). Lactate 3.2 does not fully explain.
7. Delta ratio	$(28 - 12) / (24 - 12) = 1.33 \rightarrow$ HAGMA component.
8. Oxygenation & integration	$\text{P/F} = 70/1.0 = 70$. $\text{OI} = (17 \times 1.0 \times 100)/70 = 24.3$. Severe ARDS (PALICC OI ≥ 16).

SID / Stewart - answers

Tool	Value	Interpretation
SID	$138 - 98 = 40$	High-normal - no chloride-driven component.
Atot	Albumin 4.1	Normal.
Unmeasured anions	$28 - 3.2 = 24.8$	Markedly raised - hypoxic-ischaemic anions.
Mechanism	-	Unmeasured anions + primary respiratory acidosis.

Case-specific questions

1. Identify all components of the mixed acid-base disorder.
2. Calculate the P/F ratio and oxygenation index using the available data.
3. How would you grade the severity of oxygenation failure?

Answers

1. (1) Primary metabolic acidosis (HCO₃ 12, AG 28 - hypoxic-ischaemic/lactic) and (2) primary respiratory acidosis (PaCO₂ 65, far exceeding expected ~ 26) - combined metabolic + respiratory acidosis.
2. $\text{P/F} = 70/1.0 = 70$. $\text{OI} = (17 \times 1.0 \times 100)/70 = 24.3$.
3. OI 24.3 = severe oxygenation failure (PALICC OI ≥ 16 = severe).

Post-intervention gas

Intervention. Increase ventilation, treat hyperkalaemia, fluids, consider bicarbonate.

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.25	40	17	90	138
K	Cl	Albumin	Glucose	Lactate
4.5	100	4.1	180	2.5

Post-intervention interpretation

Step	Answer
9. SID / Stewart	SID = 138 - 100 = 38 (normal). Unmeasured anions = AG (138-100-17 = 21) - 2.5 = 18.5 (still raised).

Take-home

When PaCO₂ and HCO₃ are both markedly abnormal in the same direction, name both disorders explicitly - one is not compensating for the other. SID confirms no chloride-driven component.

Case 7 - Septic shock + severe anaemia

Vignette. 4-year-old boy with septic shock and severe anaemia. FiO2 40%, SpO2 99%, ScvO2 42%.

Initial gas

pH	PaCO2	HCO3	PaO2	Na
7.25	26	11	110	138
K	Cl	Albumin	Hb	Lactate
4.3	106	4.1	4.5	7

Systematic interpretation - answers

Step	Answer
1. Clinical history	Septic shock with severe anaemia (Hb 4.5) - two independent reasons for impaired oxygen delivery.
2. Internal validity	$[H^+] = 24 \times 26 / 11 = 57 \rightarrow$ pH 7.25 - matches.
3. pH	7.25 - acidaemia.
4. Primary disorder	HCO3 11 (low) $\rightarrow$ metabolic acidosis.
5. Expected compensation	Winter's: $PaCO_2 = 1.5 \times 11 + 8 = 24.5 \pm 2$. Actual 26 - appropriate.
6. Anion gap	$AG = 138 - (106 + 11) = 21$ (raised). Albumin normal, negligible correction.
7. Delta ratio	$(21 - 12) / (24 - 11) = 0.69 \rightarrow$ mixed HAGMA + NAGMA.
8. Oxygenation & integration	$P/F = 110 / 0.4 = 275$. ScvO2 42% - tissues extracting far more O2 than usual. $CaO_2 = (1.34 \times 4.5 \times 0.99) + (0.003 \times 110) = 6.3$ mL/dL.

SID / Stewart - answers

Tool	Value	Interpretation
SID	$138 - 106 = 32$	Low - chloride-driven component from resuscitation fluids.
Atot	Albumin 4.1	Normal.
Unmeasured anions	$21 - 7 = 14$	Raised - lactic acidosis.
Mechanism	-	Two forces: low SID + unmeasured anions.

Case-specific questions

1. Calculate the P/F ratio.
2. How do you interpret the low ScvO2 despite a normal SpO2?
3. What do the severe anaemia and elevated lactate indicate about oxygen delivery and extraction?

Answers

1. $110 / 0.4 = 275$ - mild oxygenation impairment; not where the problem lies.
2. Large veno-arterial oxygen extraction gap: lungs saturating haemoglobin normally, but tissues extracting abnormally high fraction because overall DO2 inadequate relative to demand. SpO2 alone misses this.
3. Severe anaemia critically reduces CaO2 and therefore DO2, even though saturation normal. Body compensates initially by increasing extraction (low ScvO2), but once exceeded, anaerobic metabolism follows - lactate 7. Content/delivery problem, not lung problem. Priority: transfusion.

Post-intervention gas

Intervention. Blood transfusion, antibiotics, fluids, vasopressors.

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.35	36	20	100	138
K	Cl	Albumin	Hb	Lactate
4.0	105	3.8	9.0	2.5
ScvO ₂				
65%				

Post-intervention interpretation

Step	Answer
9. SID / Stewart	SID = 138 - 105 = 33 (low-normal). Unmeasured anions = AG (138-105-20 = 13) - 2.5 = 10.5 (normalising).

Take-home

A normal SpO₂ says nothing about oxygen delivery when haemoglobin is this low - look at ScvO₂ and lactate, not the saturation monitor. SID reveals the chloride-driven component.

Case 8 - Methaemoglobinaemia

Vignette. 10-year-old boy with cyanosis, headache, tachycardia, and lethargy after receiving topical benzocaine. Blood sample drawn looks chocolate-brown. FiO_2 90%, SpO_2 85%, SO_2 65%.

Initial gas

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.36	38	21	550	138
K	Cl	Albumin	MetHb	Lactate
4.3	105	4.1	35%	2

Systematic interpretation - answers

Step	Answer
1. Clinical history	Cyanosis, headache, tachycardia, lethargy after topical benzocaine, with chocolate-brown blood - close to diagnostic before numbers read.
2. Internal validity	$[\text{H}^+] = 24 \times 38 / 21 = 43 \rightarrow \text{pH } 7.36$ - matches. Essentially normal acid-base gas.
3. pH	7.36 - essentially normal.
4. Primary disorder	No significant primary acid-base disorder. Acid-base is distraction.
5. Expected compensation	Not applicable.
6. Anion gap	$\text{AG} = 138 - (105 + 21) = 12$ (normal).
7. Delta ratio	Not applicable.
8. Oxygenation & integration	$\text{P/F} = 550/0.9 = 611$ (remarkably high). Yet SpO_2 85%, SO_2 65% - large discordant saturation gap. MetHb 35% (normal <1-2%).

SID / Stewart - answers

Tool	Value	Interpretation
SID	$138 - 105 = 33$	Low-normal - mild chloride-driven component.
Atot	Albumin 4.1	Normal.
Unmeasured anions	$12 - 2 = 10$	Normal. Acid-base status benign.
Mechanism	-	Acid-base is not the story; oxygenation is.

Case-specific questions

1. Calculate the P/F ratio.
2. How do you explain the discrepancy between the very high PaO₂, SpO₂, and SO₂?
3. What is the significance of the chocolate-brown blood and MetHb 35%?
4. What is the most likely diagnosis?

Answers

1. $550 / 0.9 \sim 611$.
2. PaO₂ measures dissolved oxygen tension and is unaffected by methaemoglobin, so remains high. Methaemoglobin (Fe³⁺, unable to bind oxygen) distorts pulse oximeter algorithm, driving SpO₂ toward falsely fixed value ~82-85%. Directly measured SO₂ by co-oximetry (65%) reflects true, much lower functional saturation - 35% of haemoglobin cannot carry oxygen at all.
3. Chocolate-brown blood is classic bedside sign. MetHb 35% confirms severe elevation - enough to cause significant functional anaemia (only 65% of haemoglobin can carry oxygen). Explains cyanosis, headache, tachycardia, lethargy despite high PaO₂.
4. Acquired methaemoglobinaemia (benzocaine-induced).

Post-intervention gas

Intervention. Methylene blue 1-2 mg/kg IV (check G6PD status first), oxygen, stop benzocaine.

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.38	38	23	200	138
K	Cl	Albumin	MetHb	Lactate
4.0	105	4.1	5%	1.5
SpO ₂	SO ₂			
98%	95%			

Post-intervention interpretation

Step	Answer
9. SID / Stewart	SID = 138 - 105 = 33 (low-normal). Unmeasured anions = AG (138-105-23 = 10) - 1.5 = 8.5 (normal).

Take-home

A high PaO₂ does not guarantee oxygen-carrying capacity - whenever SpO₂ and PaO₂ disagree, request co-oximetry before you do anything else.

Case 9 - Two encephalopathic neonates

Vignette. Baby A: 4 days old, fed well until day 2, now lethargic and tachypnoeic. Afebrile, clear chest. On antibiotics for sepsis. Baby B: 7 days old, vomiting, lethargic, deep rapid breathing. Platelets 60,000/uL, urine ketones 3+.

Initial gas

A pH	A PaCO ₂	A HCO ₃	A SBE	A Na/Cl
7.51	26	20	-1	138/104
A Glucose	A Lactate	B pH	B PaCO ₂	B HCO ₃
85	1.8	7.14	18	6
B SBE	B Na/Cl	B Glucose	B Lactate	
-21	136/100	45	3.0	

Systematic interpretation - answers

Step	Answer
1. Clinical history	A: fed well until day 2, now lethargic and tachypnoeic; sepsis label but afebrile, clear chest. B: vomiting, lethargic, deep rapid breathing, thrombocytopenia, ketonuria.
2. Internal validity	A: $[H^+] = 24 \times 26 / 20 = 31 \rightarrow$ pH 7.51 - matches. B: $[H^+] = 24 \times 18 / 6 = 72 \rightarrow$ pH 7.14 - matches.
3. pH	A: 7.51 - alkalaemia. B: 7.14 - acidaemia.
4. Primary disorder	A: PaCO ₂ 26 (low) $\rightarrow$ respiratory alkalosis. B: HCO ₃ 6 (low) $\rightarrow$ metabolic acidosis.
5. Expected compensation	A: acute resp alkalosis expected HCO ₃ $\sim$ 21.2; actual 20 - appropriate. B: Winter's predicted PaCO ₂ 17 ± 2 ; actual 18 - appropriate.
6. Anion gap	A: AG = $138 - (104 + 20) = 14$ (normal). B: AG = $136 - (100 + 6) = 30$ (markedly raised).
7. Delta ratio	A: not applicable. B: $(30 - 12)/(24 - 6) = 1.0 \rightarrow$ HAGMA.
8. Oxygenation & integration	A: sepsis alone does not explain respiratory alkalosis - suspect hyperammonaemia. B: lactate only 3.0 - rest of gap filled by ketones and organic acids.

SID / Stewart - answers

Tool	Value	Interpretation
A SID	$138 - 104 = 34$	Low-normal.
A Unmeasured	$14 - 1.8 = 12.2$	Normal. Mechanism: respiratory alkalosis.
B SID	$136 - 100 = 36$	Normal.
B Unmeasured	$30 - 3 = 27$	Markedly raised - organic acids/ketones.

Case-specific questions

1. Name the primary disorder in each baby. Is compensation appropriate in each?
2. Which features does sepsis alone fail to explain?
3. Baby B has an anion gap of 30 but lactate is only 3. What fills the rest?
4. Intervene: what one test do you send in both, how must the sample be handled, and what do you do while waiting?

Answers

1. A: respiratory alkalosis, appropriate. B: metabolic acidosis, appropriate.
2. A: respiratory alkalosis without fever or distress. B: severe HAGMA with ketones and hypoglycaemia.
3. Ketones (urine 3+) and organic acids from inborn error of metabolism.
4. Ammonia. Sample on ice, delivered to lab within 15 minutes. While waiting: stop protein, give dextrose, prepare ammonia scavengers.

Post-intervention gas

Intervention. Both: ammonia level, stop protein, dextrose. A: ammonia scavengers if high. B: carnitine, treat acidosis.

A pH	A PaCO ₂	A HCO ₃	A Na/Cl	A Glucose
7.45	35	22	138/105	90
A Lactate	B pH	B PaCO ₂	B HCO ₃	B Na/Cl
1.5	7.30	25	12	138/102
B Glucose	B Lactate			
100	2.0			

Post-intervention interpretation

Step	Answer
9. SID / Stewart	A: SID = 138 - 105 = 33; unmeasured anions normal. B: SID = 138 - 102 = 36; unmeasured anions = AG (138-102-12 = 24) - 2 = 22 (still raised).

Take-home

Neonatal encephalopathy + abnormal gas = metabolic screen; ammonia on ice is critical. SID helps distinguish chloride-driven from unmeasured-anion-driven acidosis.

Case 10 - HUS-like

Vignette. 3-year-old, 5 days after bloody diarrhoea, now oliguric and pale. Hb 6.2 g/dL, platelets 40,000/uL, creatinine 3.8 mg/dL.

Initial gas

pH	PaCO ₂	HCO ₃	SBE	Na
7.22	25	10	-16	131
K	Cl	Albumin	Phosphate	iCa
6.8	104	3.0	9.5	0.92

Systematic interpretation - answers

Step	Answer
1. Clinical history	Bloody diarrhoea followed by oliguria and pallor, with anaemia, thrombocytopenia, and renal failure - HUS-like picture.
2. Internal validity	$[H^+] = 24 \times 25 / 10 = 60 \rightarrow$ pH 7.22 - matches.
3. pH	7.22 - acidaemia.
4. Primary disorder	HCO ₃ 10 (low) $\rightarrow$ metabolic acidosis.
5. Expected compensation	Winter's: $PaCO_2 = 1.5 \times 10 + 8 = 23 \pm 2$. Actual 25 - appropriate.
6. Anion gap	$AG = 131 - (104 + 10) = 17$. Albumin-corrected $AG = 17 + 2.5 \times (4 - 3.0) = 19.5$ (raised).
7. Delta ratio	$(17 - 12) / (24 - 10) = 5/14 = 0.36 \rightarrow$ mixed NAGMA + HAGMA.
8. Oxygenation & integration	Mixed acidosis; hyperkalaemia and hypocalcaemia are immediate threats.

SID / Stewart - answers

Tool	Value	Interpretation
SID	$131 - 104 = 27$	Markedly low - chloride-driven.
Atot	Albumin 3.0 (alkalinising), phosphate 9.5 (markedly acidifying)	Net high Atot.
Unmeasured anions	$19.5 - 0 = 19.5$	Raised - urate, oxalate, retained organic anions.
Mechanism	-	Three forces: low SID + high Atot (phosphate) + unmeasured anions.

Case-specific questions

1. Is compensation appropriate? What are the anion gap and the albumin-corrected anion gap?
2. Calculate the delta ratio. What does it tell you?
3. Using SID and Atot, name the three separate forces acidifying this child.
4. Intervene: which two sidekick values change your next 10 minutes? The team asks for bicarbonate - what do you say?

Answers

1. Appropriate. AG 17, corrected AG 19.5.
2. 0.36 $\rightarrow$ mixed NAGMA + HAGMA.
3. Low SID (chloride-driven), high Atot (phosphate), unmeasured anions.
4. K 6.8 (hyperkalaemia) and iCa 0.92 (hypocalcaemia). Bicarbonate? No - pH 7.22 not severe; bicarbonate can worsen hypocalcaemia and hyperkalaemia.

Post-intervention gas

Intervention. Dialysis, treat hyperkalaemia and hypocalcaemia.

pH	PaCO ₂	HCO ₃	Na	K
7.35	35	18	135	4.5
Cl	Albumin	Phosphate	iCa	
105	3.2	6.0	1.0	

Post-intervention interpretation

Step	Answer
9. SID / Stewart	SID = $135 - 105 = 30$ (improving). Atot: phosphate improved from 9.5 to 6.0. Unmeasured anions = AG ($135 - 105 - 18 = 12$) - 0 = 12 (normalising).

Take-home

HUS = mixed acidosis; hyperkalaemia and hypocalcaemia are immediate threats. SID, Atot, and unmeasured anions each tell a different part of the story.

Case 11 - Hepatitis A encephalopathy

Vignette. 6-year-old with hepatitis A, grade II encephalopathy. INR 4.2, ammonia 180 umol/L.

Initial gas

pH	PaCO ₂	HCO ₃	SBE	Na
7.49	28	21	-1	132
K	Cl	Albumin	Lactate	Glucose
3.0	98	2.2	5.5	45

Systematic interpretation - answers

Step	Answer
1. Clinical history	Hepatitis A with grade II encephalopathy, coagulopathy, hyperammonaemia.
2. Internal validity	$[H+] = 24 \times 28 / 21 = 32 \rightarrow$ pH 7.49 - matches.
3. pH	7.49 - alkalaemia.
4. Primary disorder	PaCO ₂ 28 (low) $\rightarrow$ respiratory alkalosis.
5. Expected compensation	Acute resp alkalosis: HCO ₃ falls ~ 2 per 10 mmHg fall in PaCO ₂ . PaCO ₂ 28 is 12 below 40 $\rightarrow$ expected HCO ₃ ~ 21.6 . Actual 21 - appropriate.
6. Anion gap	AG = $132 - (98 + 21) = 13$. Corrected AG = $13 + 2.5 \times (4 - 2.2) = 17.5$ (HAGMA masked by hypoalbuminaemia).
7. Delta ratio	Using corrected AG: $(17.5 - 12) / (24 - 21) = 5.5 / 3 = 1.83 \rightarrow$ mixed.
8. Oxygenation & integration	Respiratory alkalosis + masked HAGMA + hypoglycaemia.

SID / Stewart - answers

Tool	Value	Interpretation
SID	$132 - 98 = 34$	Low-normal.
Atot	Albumin 2.2	Markedly low - alkalinising, masking HAGMA.
Unmeasured anions	$17.5 - 5.5 = 12$	Raised.
Mechanism	-	Masked HAGMA + respiratory alkalosis.

Case-specific questions

1. What is the primary disorder, and why is this child alkaline?
2. Calculate the anion gap, then correct it for albumin. What changes?
3. The base excess is -1, which looks normal. Why is that misleading? How many processes are running at once?
4. Intervene: which two sidekicks need treating in the next few minutes, and why does one of them worsen the encephalopathy?

Answers

1. Respiratory alkalosis due to hyperventilation from ammonia stimulating respiratory centre.
2. AG 13, corrected AG 17.5. Low albumin was masking a HAGMA.
3. Yes - because HAGMA is masked by hypoalbuminaemia. Multiple processes: respiratory alkalosis, HAGMA (lactic acidosis), hypoglycaemia.
4. Glucose 45 (hypoglycaemia) and lactate 5.5 (lactic acidosis). Hypoglycaemia worsens encephalopathy.

Post-intervention gas

Intervention. Dextrose, lactulose, ammonia scavengers.

pH	PaCO ₂	HCO ₃	Na	K
7.42	32	20	135	3.5
Cl	Albumin	Lactate	Glucose	
100	2.5	3.0	100	

Post-intervention interpretation

Step	Answer
9. SID / Stewart	SID = 135 - 100 = 35 (normal). Atot still low (albumin 2.5) - still masking. Unmeasured anions = corrected AG (15 + 3.75 = 18.75) - 3 = 15.75 (still raised).

Take-home

Hepatitis A encephalopathy = respiratory alkalosis + HAGMA; check glucose. Always correct the AG for albumin, and check SID - the SBE may look normal while a HAGMA is masked.

Case 12 - Fasting hypoglycaemia

Vignette. 18-month-old, previously well, fasted overnight after gastroenteritis and presented with a seizure. Liver palpable 3 cm. ALT 180 U/L, ammonia 110 umol/L, free fatty acids raised.

Initial gas

Glucose	pH	PaCO ₂	HCO ₃	SBE
30	7.36	36	20	-4
Na/Cl 138/105	Lactate 2.0	b-OHB 0.3		

Systematic interpretation - answers

Step	Answer
1. Clinical history	Fasting hypoglycaemia with seizure, hepatomegaly, raised ALT and ammonia, raised free fatty acids - key is the inappropriately low ketones.
2. Internal validity	$[H+] = 24 \times 36 / 20 = 43 \rightarrow$ pH 7.36 - matches.
3. pH	7.36 - normal.
4. Primary disorder	HCO ₃ 20 (slightly low), SBE -4 $\rightarrow$ mild metabolic acidosis.
5. Expected compensation	Not applicable (primary metabolic).
6. Anion gap	$AG = 138 - (105 + 20) = 13$ (normal).
7. Delta ratio	Not applicable (NAGMA).
8. Oxygenation & integration	Near-normal gas but hypoglycaemia with low ketones $\rightarrow$ failure of ketogenesis.

SID / Stewart - answers

Tool	Value	Interpretation
SID	$138 - 105 = 33$	Low-normal - mild chloride-driven component.
Atot	Albumin not given	-
Unmeasured anions	$13 - 2 = 11$	Normal.
Mechanism	-	Acidosis mild and chloride-driven; the story is ketogenesis failure.

Case-specific questions

1. The gas looks almost normal. Why is that the most worrying finding?
2. What should a fasting child gas and ketones look like?
3. Free fatty acids are high but ketones are low. What does that mean? How would hyperinsulinism differ?
4. Intervene: list the critical sample. When exactly must it be drawn, and what do you give immediately afterwards?

Answers

1. In a fasting child, expected ketosis and metabolic acidosis. Normal gas with low b-OHB (0.3) indicates failure of ketogenesis.
2. Metabolic acidosis with high anion gap due to ketones.
3. Block in fatty acid oxidation or ketogenesis (e.g., MCAD deficiency). Hyperinsulinism would have low FFA and low ketones.
4. Must be drawn during hypoglycaemia before treatment. Includes glucose, insulin, C-peptide, cortisol, growth hormone, ketones, free fatty acids, acylcarnitine profile, ammonia, lactate, LFTs. Give dextrose (2-3 mL/kg of 10% dextrose) immediately after.

Post-intervention gas

Intervention. Critical sample, then dextrose.

Glucose	pH	PaCO ₂	HCO ₃	SBE
100	7.38	38	22	-1
Na/Cl	Lactate	b-OHB		
138/105	1.5	0.2		

Post-intervention interpretation

Step	Answer
9. SID / Stewart	SID = 138 - 105 = 33 (low-normal). Unmeasured anions = AG (138-105-22 = 11) - 1.5 = 9.5 (normal).

Take-home

Fasting hypoglycaemia with low ketones = think fatty acid oxidation defect; critical sample before treatment. SID confirms the acidosis was mild and chloride-driven, not the primary problem.

Case 13 - Shocked 2-week-old boy

Vignette. 2-week-old boy with 3 days of vomiting and poor feeding, now mottled and lethargic. Genitalia look normal. Weight below birth weight.

Initial gas

pH	PaCO ₂	HCO ₃	SBE	Na
7.27	28	12.5	-13	122
K	Cl	Glucose		
7.8	98	40		

Systematic interpretation - answers

Step	Answer
1. Clinical history	Neonatal shock with vomiting, poor feeding, mottling, lethargy, weight loss.
2. Internal validity	$[H^+] = 24 \times 28 / 12.5 = 54 \rightarrow$ pH 7.27 - matches.
3. pH	7.27 - acidaemia.
4. Primary disorder	HCO ₃ 12.5 (low) $\rightarrow$ metabolic acidosis.
5. Expected compensation	Winter's: $PaCO_2 = 1.5 \times 12.5 + 8 = 26.75 \pm 2$. Actual 28 - appropriate.
6. Anion gap	$AG = 122 - (98 + 12.5) = 11.5$ (normal). NAGMA.
7. Delta ratio	Not applicable.
8. Oxygenation & integration	NAGMA with hyponatraemia, hyperkalaemia, hypoglycaemia $\rightarrow$ CAH salt-wasting crisis.

SID / Stewart - answers

Tool	Value	Interpretation
SID	$122 - 98 = 24$	Markedly low - chloride-driven acidosis.
Atot	Albumin not given	-
Unmeasured anions	$11.5 - 0 = 11.5$	Normal.
Mechanism	-	Chloride-driven NAGMA.

Case-specific questions

1. Classify the acid-base disorder and check compensation. Is this a HAGMA or a NAGMA?
2. Three sidekick values make the diagnosis before any specialised test. Which, and what is it?
3. A vomiting 2-week-old with pyloric stenosis would show which gas instead?
4. Intervene: what are your first three actions, and which sample must precede them?

Answers

1. NAGMA with appropriate compensation.
2. Na 122 (hyponatraemia), K 7.8 (hyperkalaemia), glucose 40 (hypoglycaemia) $\rightarrow$ CAH.
3. Metabolic alkalosis (hypochloraemic, hypokalaemic).
4. (1) IV access, draw critical sample (electrolytes, glucose, cortisol, ACTH, 17-OHP, renin, aldosterone). (2) Give stress-dose hydrocortisone. (3) IV fluids with dextrose and sodium (normal saline bolus, then D5NS; no potassium initially due to hyperkalaemia). Sample must precede hydrocortisone.

Post-intervention gas

Intervention. Hydrocortisone, fluids with dextrose, treat hyperkalaemia.

pH	PaCO ₂	HCO ₃	SBE	Na
7.35	35	18	-5	135
K	Cl	Glucose		
5.0	100	90		

Post-intervention interpretation

Step	Answer
9. SID / Stewart	SID = 135 - 100 = 35 (normalising). Unmeasured anions = AG (135-100-18 = 17) - 0 = 17 (raised - resolving).

Take-home

Neonatal shock + NAGMA + hyperkalaemia + hyponatraemia = CAH until proven otherwise. SID confirms the chloride-driven mechanism and tracks response to therapy.